Independent University, Bangladesh

CSE313 · Every True / False statement, most important first

Tier 1 is asked in every past paper. Write T or F and a one-line reason for each; the reason earns the mark.

Question paper

A. Tier 1 · Which grammar works with which parser (asked in every past paper)

1. All TDP can use Left Recursive Grammar (LRG). T / F
2. All TDP can use Non-deterministic Grammar (NDG). T / F
3. Left Recursive Grammar (LRG) can not be used in LL(1). T / F
4. Non-deterministic Grammar (NDG) can not be used in LL(1). T / F
5. An LL(1) grammar can be left recursive. T / F
6. A grammar with a FIRST/FIRST conflict can be parsed by an LL(1) parser. T / F
7. If two rules fall in one cell of the table, the grammar is not LL(1). T / F
8. Brute Force Parser is a BUP. T / F
9. Recursive Descent Parser (RDP) is as powerful as LL(1). T / F
10. Recursive Descent Parser (RDP) uses OS stack. T / F
11. Recursive Descent Parser (RDP) can use Non-deterministic Grammar (NDG). T / F
12. A predictive parser needs backtracking. T / F
13. From LRG grammar complete First set can't be found. T / F
14. From LRG grammar complete Follow set can't be found. T / F

B. Tier 2 · Conversions and ambiguity

15. Left factoring removes left recursion. T / F
16. Left factoring changes the language of the grammar. T / F
17. Removing left recursion changes the language of the grammar. T / F
18. Converting an ambiguous NDG to a DG (left factoring) makes it unambiguous. T / F
19. Removing left recursion from an ambiguous grammar makes it unambiguous. T / F
20. An ambiguous grammar can be LL(1). T / F
21. Every LL(1) grammar is unambiguous. T / F
22. In an unambiguous expression grammar, + is defined at a higher level (closer to the start symbol) than *. T / F
23. A left recursive rule (E → E + T) makes the operator left associative. T / F
24. The rule for ^ (power) should be left recursive. T / F

C. Tier 3 · FIRST, FOLLOW and the LL(1) table

25. The FIRST set of a non-terminal can contain ε. T / F
26. The FOLLOW set of a non-terminal can contain ε. T / F
27. $ is always in FOLLOW of the start symbol. T / F
28. FIRST of a terminal is the terminal itself. T / F
29. For A → ε, the rule is placed in M[A, b] for every b in FOLLOW(A). T / F
30. ε can be a column in the LL(1) parsing table. T / F
31. An LL(1) parsing table has one row for each terminal. T / F
32. In LL(1), the first L means the input is scanned Left to right. T / F

D. Tier 4 · Parser classes, phases and lexical analysis

33. A Bottom-Up Parser builds the parse tree from the root down to the leaves. T / F
34. Bottom-up parsing uses the reverse of the rightmost derivation. T / F
35. The lexical analyzer removes comments and white space. T / F
36. The lexical analyzer builds the parse tree. T / F
37. Syntax analysis comes after lexical analysis. T / F
38. Semantic analysis belongs to the analysis (front end) part of the compiler. T / F
39. An interpreter produces a target program in machine code. T / F
40. Regular expressions can check balanced parentheses. T / F